

Education

Okinawa Institute of Science and Technology

Ph.D. in Computational Neuroscience

Okinawa, Japan

Jan. '25 — May '26

- Withdrew from the program in good standing to better pursue personal research interests and career goals
- Selected coursework: Neuromotor Control, Nonlinear Dynamical Systems, Brain Computation, Cognitive Neurorobotics

University of Western Ontario

M.Sc. in Neuroscience; Grade: A

London, Canada

Sept. '22 — Oct. '24

- Thesis title: The Nature of Reflexes in Online Planning and Control [\[link\]](#)
- Recipient of the Vector Scholarship in Artificial Intelligence (C\$17.5k) and the BrainsCAN Graduate Studentship (C\$50k)
- Selected coursework: Principles of Neuroscience, Reinforcement Learning

University of Toronto

B.A.Sc. in Electrical Engineering; cGPA: 3.92/4.0

Toronto, Canada

Sept. '17 — Apr. '22

- Engineering International Scholar: received a full tuition fee waiver for the entire duration of the program (C\$229k)
- Recipient of the Adel S. Sedra Gold Medal for achieving the highest cumulative average in the graduating class
- Graduated with High Honors and Minor in Robotics and Mechatronics; Dean's Honor List in all semesters
- Selected coursework (*graduate-level): Linear Control Systems, *Random Processes, *Sensory Communication, Analog and Digital Electronics, System Mapping, Machine Learning, Real-time Control, Robot Modeling and Control, Mechatronics

Work Experience

Doctoral Researcher, OIST Graduate University

Biological Physics Theory Unit and Information Theory, Probability, and Statistics Unit

Okinawa, Japan

Jan. '25 — May '26

- Focusing on understanding closed-loop neural dynamics through behaviorally informative decomposition of control
- Completed several mini projects on topics like adaptive output chunking in recurrent neural networks, passive-active optimal control, dynamical systems reconstruction, speed-curvature power-law, and I/O stability of plastic spiking neural networks

Graduate Researcher, Western Institute for Neuroscience

Sensorimotor Superlab

London, Canada

Sept. '22 — Oct. '24

- Investigated goal-dependent coupling between reflexive and voluntary motor control by estimating update times of both modalities independently during ongoing movement; also tried investigating role of motor variability in implicit adaptation
- Led multiple discussions on motor planning and control; also mentored and trained undergraduate research assistants

Navigation Engineering Intern, Zebra Technologies

Path-Planning and Control Team

Mississauga, Canada

May '20 — Aug. '21

- Critiqued, implemented, and tested auto-navigation algorithms from literature for deployment on inventory scanning robots
- Redesigned path planner to improve aisle scan coverage and efficiency by robustly handling obstructions and curved aisles
- Developed tools for rapidly prototyping planners and controllers and benchmarking their performance; written in Julia
- Reviewed and fixed real-world behavior and performance bugs; identified, proposed, and applied planner improvements

Software Engineering Intern, Rocscience Inc.

Geotechnical Software Tools Design

Toronto, Canada

May '19 — Aug. '19

- Developed interactive geometry import and diagnostic tools for file formats external to, and also within, the software suite

Undergraduate Research Assistant, Motion Adaptation Science Lab

Toronto Rehabilitation Institute

Toronto, Canada

May '18 — Aug. '18

- Created finite element models of the lower leg from MRI data for optimizing functional electro-stimulation strategies

Technical Interests

Neuro-inspired Robotics · Systems ID and Integration · Generalizable Control · Perception · Embodied Intelligence

Skills

Python · Julia · Matlab · C/C++ · ROS · PyTorch · OpenCV · Typst · Graphics Design · Mechatronics · Machining

Professional Activities

Conference Poster (CRSy, 2025) · Invited Talk (IIT Gandhinagar, UC Louvain, 2024) · Student Leadership & Service